Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/26/2008
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Vigilance, Stress and Sleep/Wake Measures and Low Autonomy Study in NEEMO 13-A Simulated Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2007	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NBPF00803		
Performance Goal No.:			
Performance Goal Text:			
	NASA Extreme Environment Mission Operations (NEEMO) has astronauts live in the NOAA Aquarius habitat (19 meters underwater off the coast of Florida). Aquarius is an excellent space analog environment for a number of challenges in the area of behavior and performance: confinement, limited personal space, isolation, workload timeline demands, EVAs, environmental risk, no quick return to surface in an emergency. Aquarius is operated by the National Underwater Research Center of the UNC at Wilmington. NEEMO missions work best for research that is directed toward proof of concept and feasibility studies in isolated and extreme environments. Research that needs testing in an analog mission works well in a NEEMO setting as it gives the researcher an idea of how the product works in a confined, but crowded environment focused on multiple mission accomplishments. The 11-day NEEMO 13 mission—from August 6-15, 2007, was co-sponsored by NASA and the National Space Biomedical Research Institute (NSBRI). The primary operational objectives of NEEMO 13 focused on testing lunar		

Task Description:	exploration concepts, autonomy, and a suite of long-duration spaceflight medical objectives. The 4-person crew conducted a variety of undersea "moon walks" to test concepts for future lunar exploration using advanced navigation and communication equipment. The medical objectives involved behavioral health and performance, immune function changes, oxidative damage and hardware and/or software evaluations of Portable Unit of Metabolic Analysis (PUMA) and GuideView. The crew worked much more independently from the mission control team than on previous missions. The autonomous mode of operation encouraged the crew to make real-time decisions about daily operations similar to what we think will be necessary for lunar and Mars missions. The approach is to show how procedures and training for future missions can be adapted, considering the reduced direct communication with Mission Control those crews will encounter. This report convers specifically the behavioral health and performance measures in NEEMO 13 that involved testing the usefulness of the 3-minute Psychomotor Vigilance Test of performance; acquisition of salivary cortisol measures of stress; and (3) monitoring sleep-wake patterns using ambulatory technology (wrist actigraphs and sleep logs). Data acquisition in NEEMO 13 was successful. A total of 149 PVT performance tests were acquired, 150 salivary cortisol measures were obtained, and 36 days of wrist actigraphy, sleep-wake diary and subjective fatigue and stress scales were acquired, for a 97% adherence rate. The salivary cortisol and subjective ratings revealed no elevated stress during NEEMO 13, and the wrist actigraphy, sleep-wake diary, and subjective fatigue data from NEEMO 13 revealed no sleep loss or elevated fatigue. These outcomes made the 3-minute PVT performance data from NEEMO 13 useful for adding to the normative PVT performance database for astronauts begun with PVT data from NEEMO 9 and 12 missions. The resulting larger 3-minute PVT performance database on N=10 astronauts and 2 aquanauts will be used to generate a PVT SelfTest user interface for use by astronauts in space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of a 3-minute PVT performance test that is a very sensitive objective measure of fatigue and sleepiness in astronauts during space flight, will also have extensive utility in a wide range of activities on Earth, including evaluating readiness to perform in military personnel, tarnsportation and secutiry personnel and people working in high risk operational environments (e.g., power plants), as well as use in identifying excessive sleepiness in patients.
Task Progress:	During NEEEMO 13, a total of 149 PVT performance tests were acquired, 150 salivary cortisol measures were obtained, and 36 days of wrist actigraphy, sleep-wake diary and subjective fatigue and stress scales were acquired, for a 97% adherence rate. The salivary cortisol and subjective ratings revealed no elevated stress during NEEMO 13, and the wrist actigraphy, sleep-wake diary, and subjective fatigue data from NEEMO 13 revealed no sleep loss or elevated fatigue. These outcomes made the 3-minute PVT performance data from NEEMO 13 useful for adding to the normative PVT performance database for astronauts begun with PVT data from NEEMO 9 and 12 missions. The resulting larger 3-minute PVT performance database on N=10 astronauts and 2 aquanauts will be used to generate a PVT SelfTest user interface for use by astronauts in space flight.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Awards	Dinges DF. "Laurence R. Young Space Biomedical Research Award for contributions to human performance in space, February 2008." Feb-2008